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EMERGENCE AS NON-AGGREGATIVITY AND THE BIASES OF REDUCTIONISMS*

ABSTRACT. Most philosophical accounts of emergence are incompatible with reduction. Most scientists regard a system property as emergent relative to properties of its parts if it depends upon their mode of organization—a view consistent with reduction. Emergence is a failure of aggregativity, in which “the whole is nothing more than the sum of its parts”. Aggregativity requires four conditions, giving powerful tools for analyzing modes of organization. Differently met for different decompositions of the system, and in different degrees, the structural conditions can provide evaluation criteria for choosing decompositions, “natural kinds”, and detecting functional localization fallacies, approximations, and various biases of vulgar reductionisms. This analysis of emergence and use of these conditions as heuristics is consistent with a broader reductionistic methodology.

KEY WORDS: additivity, aggregativity, emergence, functional localization fallacies, heuristics, near-decomposeability, reduction, whole-parts relations

I. INTRODUCTION: REDUCTION AND EMERGENCE

Philosophers commonly suppose that emergent properties are irreducible, but some rather nice things fall out of a reductive account of emergence. Claims involving emergent properties in discussions of non-linear dynamics, connectionist modelling, chaos, artificial life, and elsewhere give no support for traditional anti-reductionism or woolly-headed anti-scientism. But reductionists often misinterpret their success, claiming more than they can possibly deliver. Emergent phenomena like those discussed here are often subject to surprising and revealing reductionistic explanations.¹ But such explanations do not deny their importance or make them any less emergent – quite the contrary: it explains why and how they are important, and ineliminably so. This requires a broader view of reduction (and emergence) than common among philosophers,



Foundations of Science 5: 269–297, 2000.

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but one scientists would recognize.² *A reductive explanation of a behavior or a property of a system is one which shows it to be mechanistically explicable in terms of the properties of and interactions among the parts of the system* (See also Kauffman, 1971).³ The relevant explanations are causal, but needn't be deductive or involve laws – contrary to conventional wisdom (Wimsatt 1976b; Cartwright 1983).

This account analyzes emergence in terms of dependence of a system property on the arrangements of the parts – ultimately therefore on the *context-sensitivity* of relational parts' properties *to intra-systemic conditions*. Some accounts of emergence suppose *extra-systemic context-sensitivity* of system properties.⁴ Neither need be anti-reductionistic. The latter would require finding a larger embedding system including the initially extra-systemic properties engaging the broader context-sensitivities. If an adequate mechanistic account could then be provided, a reductive account using the original system would have failed, but one at a higher level would have succeeded – (including the original system together with more variables from its environment).⁵

Such level and context or scope switching – a well-hidden secret of reductionistic approaches – can also be a useful strategy to remove modeling biases resulting from “perceptual focus” on objects at a preferred level. These biases can be mutually reinforcing and extremely powerful. They led to the invisibility of false simplifying assumptions about the structure of groups in models of group selection (Wade, 1978; Wimsatt 1980b). The models focussed in a plausible way on genes and individual organisms, making in the process standard simplifying assumptions appropriate for some questions at those levels, but not for almost any questions at higher levels of organization, *and equivalent to assuming that groups did not exist*. Not surprisingly, these assumptions led to the conclusion that group selection was not significant! We must work back and forth between levels to check that features crucial to phenomena at an upper level are not simplified out of existence when modelling at the lower level. On the analysis below, we would have shown that these group properties were emergent at the higher level, although consistent with a sensitive reductive analysis.

Some philosophers who defend “qualia” type accounts of subjective experience suppose a third (“mystical”) notion of emergence which – if coherent – would be anti-reductionistic. I won’t consider it here. Finally, many accounts appeal to multiple-realizeability as a criterion for emergence, and link it with failures of reduction. I have argued elsewhere (1981a, 1994) that multiple realizeability is entailed by the existence of compositional levels of organization, is much broader than commonly supposed, and neither mysterious nor contrary to reductionism.

Many cases were classically considered as involving emergence – cases motivating claims that “the whole is more than the sum of the parts.” – like an electronic oscillator circuit. There’s nothing anti-reductionistic, mysterious, or inexplicable about being an oscillator. You hook up an inductance, a capacitor, and a resistor in the right way with a voltage source. The system oscillates although none of its parts in isolation do so. Moreover, it is how these disparate parts are strung together that makes them an oscillator. (Thus, an oscillator must contain a closed circuit.) A deductive theory relates properties of the parts to the frequency and amplitude of the oscillator; reductionistic even by the strong conditions of the formal model of reduction, and also by the weaker characterization given here. This is intuitively a case of emergence, though it can’t be if we tie emergence to non-reduceability.

More generally, *emergence of a system property relative to the properties of the parts of that system indicates its dependence on their mode of organization*. It thus presupposes the system’s decomposition into parts and their properties, and its dependence is explicated via a mechanistic explanation. Ten out of twelve cases from Table I below fail one or more of the conditions for aggregativity, are consistent with totally reductionistic accounts of the systemic phenomena in question, depend in some way on the mode of organization of the parts, and are *prima facie* examples of emergence. Not every counterexample kills an analysis, but too many which are too central do. These surely qualify. We need an analysis of emergence consistent with reductionism.

I proceed indirectly by focussing on another question: When intuitively is a system “more than the sum of its parts”? This question has been classically connected with discussions of reduc-

tion and of emergence, and has stimulated a variety of erroneous but revealing views. How these views arise and may contribute to species of vulgar reductionism – “Nothing-but-isms” – through biases in reductionistic problem-solving strategies is discussed at the end. Their pervasiveness is a back-handed tribute to the importance of approximations, idealizations, and limiting case arguments in science. These are powerful tools, but like any tool, they can be misused – we must learn how to use them properly.

II. AGGREGATIVITY

Emergence should involve some kind of organizational interdependence of diverse parts, but there are many possible forms of such organizational interaction, and it is hard to know how to classify them. One can't easily turn examples of the organizational interaction of diverse parts into a general analysis of emergence. It is easier to discuss *failures* of emergence, so I proceed in a backwards fashion (Wimsatt, 1986), by determining conditions for the system property *not* to be emergent (i.e, for it to be a “mere aggregate” of its parts properties). This has a straightforward, revealing and compact analysis. Forms of emergence can then be classified and analyzed more systematically by looking at how *these* conditions can *fail* to be met. Examples for different properties and decompositions of the system into parts are given in Table I.

Four conditions seem separately necessary and jointly sufficient for *aggregativity*⁶ or non-emergence. Aggregativity and emergence concern the relationship between a property of a system under study, and properties of its parts. For each condition, the system property must remain invariant or stable under modifications of the system in the specified way, a kind of independence of the property over changes in the mode of organization of the parts. This invariance indicates that the system property is not affected by wide variation in relationships between and among the parts and their properties. To be aggregative, the system property would have to depend upon the parts' properties in a very strongly atomistic manner, under all physically possible decompositions – an almost impossibly strong demand. *It is rare indeed that all of these conditions are met* (See Table I). Aggregativity is the complete antithesis of func-

TABLE I

EXAMPLES OF FAILURES OF AGGREGATIVITY as discussed in Wimsatt, 1986 (with indications of the decompositions assumed and which conditions are met)

Examples: Invariance Conditions: R = <i>restricted or qualified</i> (for some decompositions only)	IS:	QS:	RA:	CI:	Relevant parts or conditions:
1. mass	yes	yes	yes	yes	none known. Only case of true aggregativity in table.
2. volume	R	yes	yes	no	not for solvent/solute.or packing diff. solids
3. critical mass of fissionable material. (ignoring neutron absorbers in context)	yes	no	yes	no	keep geometrically and density invariant
4. stability of rock pile (yes for IS if same = exactly similar)	no/yes	no	no	no	
5. oxygen binding in hemoglobin molecule	R	no	?	no	w/i alpha, beta chains
6. gamete frequency (classical model) (these two cases differ in unit chosen)	no	no	no	yes	units are genes
7. gamete frequency (classical model)	yes	no	no	yes	units are chromosomes (w/o disturbing mitosis)
8. gene effects (classical model) (these two cases reflect a change in theory)	yes	no	yes	yes	units are genes
9. position effect (molecular model)	no	no	no	?	units are genes
10. amplification ratio in multi-stage linear amplifier	yes	yes	yes	yes	idealization; not found in Nature (Figure 1)
11. amplification ratio in multi-stage non-linear amplifier	no	yes	no?	yes	analogous natural case (Figure 2)

TABLE I

Continued

Examples: Invariance Conditions: R = <i>restricted or qualified</i> (for some decompositions only)	IS:	QS:	RA:	CI:	Relevant parts or conditions:
12. solving a nearly-decomposeable problem	yes	yes	R	yes	units are subproblems; aggregativity usually is partial or qualified.
13. computer memory	yes	yes	R	yes	memory chips having characteristics
14. fitness (total additivity) (<i>QS is met if positive alleles are substituted for others at same locus without affecting genome size</i>) (for different groups of genes in same organism)	yes	no	R	yes	units are genes for a "quantitative trait".
15. fitness (partial additivity-classical)	no?	no	R	no	units are genes w. on-0
16. Levins' fine and coarse grained fitness functions	yes	yes	yes	yes	<i>approximations invalidate claim of aggregativity*</i>

*Appearances are deceiving: conditions of derivation violate conditions **IS**, **RA**, **CI**, even though the composition functions for both the coarse-grained and fine-grained adaptive functions appear to meet all conditions. This situation should be generalizable to many idealizations, or derivations involving approximations. (See Wimsatt, 1986 and below for details.)

(1) In addition to the advantages of breaking down claims of emergence into these 4 criteria which can be met in various combinations, thus providing a finer classification of types of emergence, there is a substantial heuristic advantage in using these criteria in the analysis of complex systems. This arises because the criteria are met usually under restricted conditions on the parts decomposition involved, and sometimes with some additional restrictions. These criteria then become heuristic templates for uncovering these conditions, and for finding parts decompositions which realize these criteria to the greatest extent. These decompositions are prime candidates for specifying "natural kinds." (2) Example #16 shows that a purely formal account in terms of conditions on the composition function will not do; one needs also to look at the assumptions made in the derivation of the composition function. (3) See also sub-aggregativity, neighborhood aggregativity, modular aggregativity, semi-aggregativity, and near-aggregativity in Wimsatt, 1986 for other kinds of qualified aggregativity.

tional organization. Our reductionistic science to date has focussed disproportionately upon such properties, or properties which do somewhat less – meeting some of these conditions, approximately, some of the time and studying them under conditions in which they are “well-behaved”. Aggregative or even such “pseudo-aggregative” properties are treated as relatively fundamental (Martinez, 1992). In consequence, their import in the description of the natural world has been substantially exaggerated.

These conditions are: (1) a condition on the intersubstitution or rearrangement of parts; (2) on size scaling (primarily, though not exclusively, for quantitative properties) with addition or subtraction of parts; (3) on invariance under decomposition and reaggregation of parts; and (4) a linearity condition that there be no cooperative or inhibitory interactions among parts in the production or realization of the system property. These conditions are not entirely independent of one another. In fact, there seem to be close connections between (1) and (3) and between (2) and (4). The conditions (Wimsatt, 1986) are as follows:

For a system property to be an aggregate with respect to a decomposition of the system into parts and their properties, the following four conditions must be met:

Suppose $P(S_i) = F\{[p_1, p_2, \dots, p_n(s_1)], [p_1, p_2, \dots, p_n(s_2)], \dots, [p_1, p_2, \dots, p_n(s_m)]\}$ is a composition function for system property $P(S_i)$ in terms of parts' properties $p_1, p_2, \dots, p_n$, of parts $s_1, s_2, \dots, s_m$. The composition function is an equation – an inter-level synthetic identity, with the lower level specification a realization or instantiation of the system property.⁷

1. **IS (*Inter Substitution*)** Invariance of the system property under operations rearranging the parts in the system or interchanging any number of parts with a corresponding numbers of parts from a relevant equivalence class of parts. (cf. commutativity of composition function).

2. **QS (*Size scaling*)** Qualitative Similarity of the system property (identity, or if a quantitative property, differing only in value) under

addition or subtraction of parts. (cf. recursive generability of a class of composition functions).

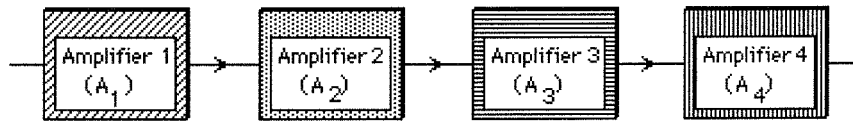
3. **RA** (Decomposition and *ReAggregation*) Invariance of the system property under operations involving decomposition and reaggregation of parts. (cf. associativity of composition function).

4. **CI** (*Linearity*) There are no **C**ooperative or **I**nhibitory interactions among the parts of the system which affect this property.

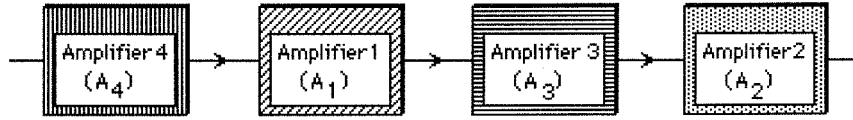
Note that conditions **IS** and **RA** are obviously relative to given parts decompositions, as are (less obviously) **QS** and **CI**. A system property may meet these conditions for some decompositions, but not for others. Table I presents different examples of aggregativity and its failure – species of emergence (Many of these examples are discussed in Wimsatt, 1986).

Figure 1 illustrates the first three conditions for the amplification ratio in an (idealized) multi-stage linear amplifier. The system property, total amplification ratio, is the product of the component amplification ratios (Figure 1a), a composition function which is commutative (Figure 1b, condition **IS**), and associative (Figure 1d, condition **RA**), and shows qualitative similarity when adding or subtracting parts (Figure 1c, condition **QS**). (This example seems to violate the 4th condition (non-linearity), but we act as if it doesn't: geometric rates of increase are treated linearly here (and in other relevant cases) because it is the exponent which is theoretically significant for the properties we are interested in.) Subjective volume grows linearly with the exponent (as our decibels scale reflects), and both of them grow linearly with addition of components to the chain. The 4th condition is found for cooperative interactions in the hemoglobin molecule: the four subunits take up and release oxygen more efficiently by being organized into a tetramer than they would as four independent units. (A monomeric hemoglobin can be found for comparison in the lamprey.) The staged linear amplifiers are interesting because they show that aggregativity does not literally mean "additivity" – here multiplicative relations do equally well.⁸ (Exponential growth is also much more common in biology than linear relations.)

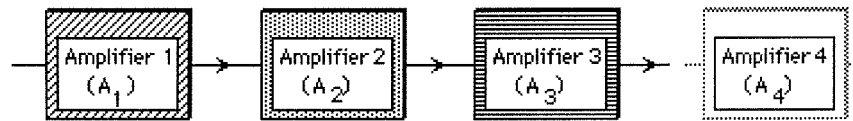
The linear amplifier of Figure 1a–d is case 10 of Table I. To meet all criteria, we must assume that each sub-amplifier is exactly linear



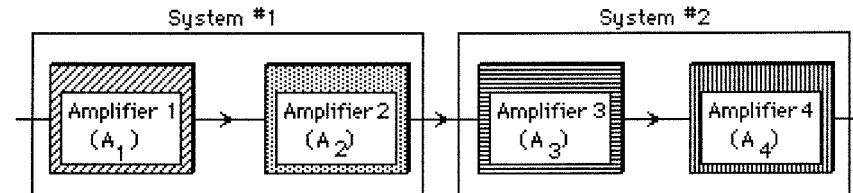
1a: Total Amplification Ratio, A_t , is the product of the amplification ratios of the individual amplifiers: $A_t = A_1 \times A_2 \times A_3 \times A_4$.



1b: Total Amplification ratio, $A_t = A_4 \times A_1 \times A_3 \times A_2$ remains unchanged over intersubstitution: changing the order of the amplifiers (or commutation of the A's in the composition function).



1c: Total Amplification Ratio, $A_t(n) = A_t(n-1) \times A(n)$, remains qualitatively similar when adding or subtracting parts.



1d: Total Amplification Ratio is invariant under subsystem aggregation—it is associative: $A_1 \times A_2 \times A_3 \times A_4 = (A_1 \times A_2) \times (A_3 \times A_4)$.

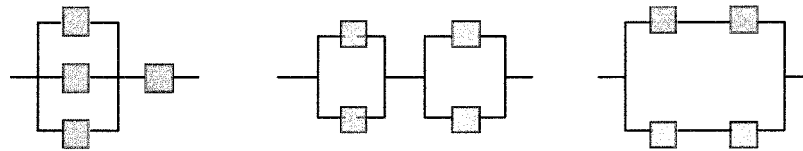


Figure 1. Conditions of Aggregativity illustrated with idealized linear unbounded amplifiers.

throughout the entire range – from the smallest input to the largest output – required of the entire system. No real-world amplifiers here: this is an idealization, which I will return to below.

Even this simple story has some important limits however. Amplifiers are themselves integrated functional wholes with differentiated parts – which cannot be permuted with impunity. (That is why we need circuit diagrams to assemble and to understand them –

we cannot put them together in just any fashion!) Even the parts are integrated wholes. If you cut randomly through a resistor or capacitor, the pieces do not perform like the original. This is interesting: *testing these conditions against different ways of decomposing the system is revealing of its organization*. This suggests broader uses for the conditions and the analysis.

We aren't done yet: all examples so far (Figures 1a–1d) have the amplifiers arranged in series. This is an implicit organizational constraint on the whole system – readily accepted because our common uses of amplifiers connect them in this way. But we could also connect them differently, as in the three series-parallel networks diagrammed in Figure 1e, and then the invariances in total amplification ratio, $\bullet A$, disappear. To calculate their amplification ratios, we make three (simplifying) assumptions: (1) branching parallel paths divide currents equally; (2) converging parallels add currents; (3) serial circuits (included aggregated sub-assemblies) multiply signal strengths (as before). In Figure 1e, starting with a signal of magnitude 1 unit, and following these rules we get:

For the first circuit, $\bullet A = [(A1 + A2 + A3)/3] \times A4$.

For the second circuit, $\bullet A = [(A1 + A2)/2] \times [(A3 + A4)/2]$

For the third circuit, $\bullet A = [(A1 \times A3)/2] + [(A2 \times A4)/2]$

If all component amplifiers, $A1, A2, A3, A4$ have equal amplification ratios, a , the $\bullet A$'s of these circuits are all equal. They each have two staged subassemblies, each with a net amplification ratio of a , so $\bullet A = a \times a = a^2$. (For these ideal linear amplifiers, parallating at any stage has no effect: the signal is just multiplied by the same amount along each path, so they collectively have the same effect as a single amplifier with the same amplification ratio.) But the decreased amplifying depth reduces $\bullet A$ from a^4 to a^2 , so changing to any of these modes from a strictly serial circuit decreases the amplification ratio. But there is worse to come:

Suppose that the amplification ratios of the component amplifiers are not equal.

If $A1 = a$, $A2 = 2a$, $A3 = 4a$, and $A4 = 8a$, then

circuit 1 = $[(7/3) a] \times 8a = 56/3 a^2 = 18.67a^2$.

circuit 2 = $(3/2) a \times 6a = 9a^2$.

circuit 3 = $(5/2) a^2 + (10/2) a^2 = 7.5a^2$.

For comparison, all of them in series (as above) would give $64a^4$. So how the amplifiers are connected together *does* make a difference. Despite first appearances, amplification ratio is not an aggregative property, both because it is not invariant across decompositions internal to the amplifiers, or internal to their parts, and also because it is not invariant across aggregations of the amplifiers which are not serially organized. (This covers manipulations at three levels of organization: rearrangements of parts of the amplifier parts, rearrangements of parts of the amplifiers, and rearrangements of the amplifiers.)

Finally, we have assumed that each sub-amplifier is exactly linear throughout the entire range from the smallest input to the largest output – required of the entire system. They must multiply input signals of different frequencies and amplitudes by the same amount over this entire range. This is an idealization – case 10 of Table I. Real-world amplifiers are *approximately* linear through given power and frequency ranges of input signals. (See Figure 2a and case 11 of Table I). (Frequency correction curves are published so that the user can restore linearity by “boosting” different frequencies by different amounts, but these curves are themselves functions of the amplitude of the input signal.) The amplifiers – not perfectly linear to begin with – become increasingly non-linear outside these ranges. They are most commonly limited on the low side by insensitivity to inputs below a certain value, and on the high side by not having enough power to keep the transformation linear. So with real amplifiers the order of the amplifiers *does* matter, even in the serial circuit.

Thirty years ago hi-fi systems had separate pre-amplifiers and amplifiers. Hooked up in the right order, they worked just fine. In the wrong order, the amplifier would be too insensitive to detect the signal from the phono-cartridge, but small amounts of “white noise” it generated internally would be amplified enough to “fry” the downstream pre-amp. So-called linear amplifiers really mean *approximately linear over a given range, within a specified tolerance, ε* . Different uses require different tolerances and ranges, so a system may be treated as linear for some purposes, but not for (more demanding) others. (See below for more on the implications of tolerances).

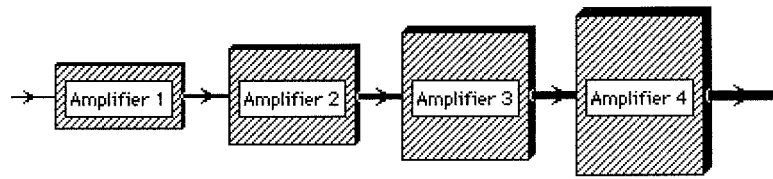


Figure 2a: Amplifiers with minimum input and maximum output signal strengths, as indicated by their relative heights. Their order is not interchangeable without introducing non-linear distortion.

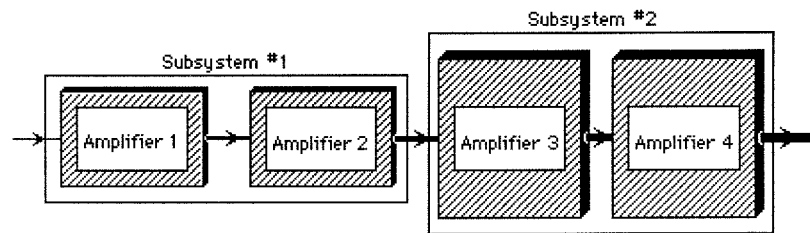


Figure 2b: partial or sub-aggregativity: The amplifiers in each of the two sub-systems are interchangeable with each other without introducing deviations from non-linearity in the overall system, but intersubstitutions between sub-systems introduces non-linear distortions.

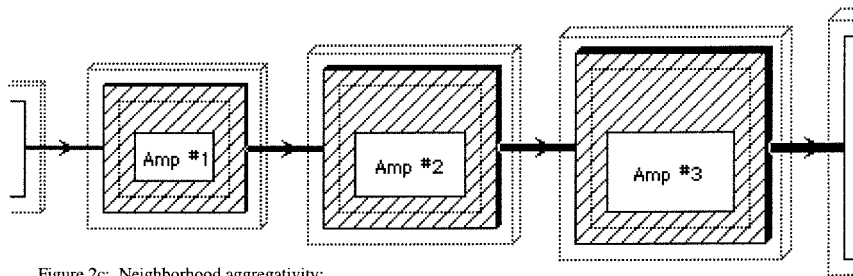


Figure 2c: Neighborhood aggregativity:
Any amplifier is intersubstitutable with its immediate neighbor to the left or to the right, but not further without introducing non-linear distortions. (Range of intersubstitution is indicated by dotted boxes inside and outside of each amplifier.)

Figure 2. Varieties of Partial Aggregativity illustrated with amplifiers which are linear only within a bounded range.

The amplifiers may be linear through a large enough range that there is some adjustability – some restricted intersubstituteability among the components, yielding “partial” or “sub”-aggregativity, (Figure 2b) or “neighborhood” aggregativity (Figure 2c). Restricted kinds of intersubstitutability (see also “modular” aggregativity, Wimsatt, 1986) can also help characterize different modes or dimensions of organization in many kinds of complex systems. Similar qualifications for supposedly aggregative properties arise in discussions of additive fitness components, and fine vs. coarse-grained

patterns of environmental change (Wimsatt, 1999). *Indeed, the appearance of common and unqualified aggregativity is a chimera – usually a product of uninspected assumed constancies, idealizations, and overlooked possible dimensions of variation.* The amplifier case demonstrates how known idealizations and uninspected assumptions (treating amplifiers and their parts as unbreakable modules because they are commonly treated as such in our theory and practice, and considering only serially organized amplifiers because these modes are common for functional reasons) can lead us to exaggerate the aggregativity of a property. ***In the analysis of complex system properties, such kinds of errors are so easy to commit that they are almost the rule rather than the exception, contributing to design failures in engineering, modelling errors and errors of experimental design in science, and conceptual errors in philosophy.***

But some properties at least seem like paradigmatic aggregative properties. The great conservation laws of physics – those of **mass** (case 1), **energy**, (now replaced by the hybrid, mass-energy), **momentum**, and **net charge**, (if we include its sign) in effect indicate that these properties actually do fill the bill. They appear aggregative under any and all decompositions. Indeed, that's why there are conservation laws for them! Curiously, some properties you might have expected to don't measure up. Thus, *volume*, (Table I, case 2) isn't aggregative, if we consider solvent-solute interactions in chemistry. If you dissolve salt in water, the volume of the water + salt will be less than that of the water before the salt was added, and surely less than their volumes taken independently and separately. (So sometimes the whole is **less** than the sum of its parts!)

A system property may be aggregative for some decompositions but not for others, or, more generally, any of the conditions may be met for some decompositions, but not for others. (This is probably most common.)

This has critical importance in theory construction: ***these variations allow for and suggest feedback between these criteria and the choice of decompositions of a system for further analysis. We look for invariances, so these conditions are treated as desiderata and we experiment with alternative descriptions of and manipulations on the system hoping to find ways to make them work –***

*decomposing, cutting, pasting, and adjusting until they are satisfied to the greatest degree possible. Furthermore, we will tend to regard decompositions better meeting aggregativity conditions as “natural”, because they provide simpler and less context-dependent regularities, theory, and mathematical models involving these aspects of their behavior.*⁹

Other central grounds for regarding decompositions and the parts produced as “natural” include ‘robustness’ (Levins, 1966; Wimsatt, 1981a), and more enigmatically, ‘generative entrenchment’ (Wimsatt, 1986). Other heuristics are also used with these conditions in constructing and validating decompositions – see the reductionistic problem solving strategies discussed in my 1980b, 1985, and 1997. These heuristics have systematic biases, which may give misleading results. One of the most systematic biases in discussions of supposedly aggregate behavior is to generate it under very special conditions or strong constraints on the system and its environment, which are then forgotten in subsequent discussions.

These four conditions are sufficiently formal in character to suppose that we could move right to an evaluation of the properties found in the equations of theories. But I show elsewhere (Wimsatt, 1999, 1986) that we can’t properly do so without considering at least (1) the choice of a parts-decomposition, (2) idealizations and assumptions made in the *description* of a system, and (3) further idealizations and approximations made in the *derivation of equations* relating system-level and parts-level properties.¹⁰ Examples #7 and #8 from Table I reveal how the apparent aggregativity of a system property depends upon the decomposition used.

Moreover *one cannot simply look at the form of an expression relating system and parts’ properties to tell whether a system property is aggregative*. You need to look also at the assumptions made in deriving the “composition function” for the system property, and make sure that all of *them* are empirically adequate for the case in question. This places important limitations on a formalistic account of aggregativity, for it is not enough to look at the equations in the finished empirically adequate theories. You also have to know how you got there, *because the approximations you made along the way cannot be forgotten in evaluating aggregativity*.

III. AGGREGATIVITY AS A HEURISTIC FOR EVALUATING DECOMPOSITIONS, AND OUR CONCEPTS OF NATURAL KINDS

These cases have yielded many interesting features of claims of aggregativity or partial aggregativity. Seeing them together suggests interactions among the development of theory, methods of decomposition, and experimental design, and what we make of what we have found. Together they give a very different picture of the nature and uses of aggregativity than commonly assumed, and have further implications for the assessment (and biases) of reductionistic methodologies.

(1) From Table I, we see that very few system properties are aggregative functions of parts' properties, suggesting that emergence – as failure of aggregativity – is extremely common; the rule, rather than the exception. (The conservation laws of physics pick out aggregative properties, but little else does.) So this analysis could – perhaps fairly – be criticized as yielding a very weak notion of emergence. Nonetheless it accords with intuitions of most scientists, who are not willing to give up either their reductionism or their emergence, and who agree with its classification of particular cases.

(2) But then one might wonder why the temptation for “Nothing but-ism” – the ontological war-cry of what Dennett (1995) calls “greedy reductionism” – is so strong. We see statements quite regularly in science like “Genes are the only units of selection”, “Organisms are nothing but bags of genes”, “The mind is nothing but neural activity”, “Social behavior is reducible to or nothing more than the behavior of individuals.” If total aggregativity is so rare, why are claims like these so common? They clearly aren't claiming full-blown aggregativity in the sense explicated here. While true aggregativity requires invariance of the system property under *all* decompositions and reaggregations, we may (fallaciously) think of behavior at higher levels as being aggregates of the behavior of parts for *particular* decompositions which *do* show this invariance – or show it only approximately. Such properties *look* aggregative for some decompositions, but reveal themselves as emergent, or context or organization-dependent for other decompositions or conditions. This is particularly clear in the discussion of multi-

locus genetic systems in the context of a theory, population genetics, which currently emphasizes the evolutionary role of additive fitness interactions, and ignores the central role of epistasis (Wade, 1996; Wimsatt, 2002, 1980b). Pointing to such practices naturally changes our focus of investigation from how to specify relations between system and parts' properties to looking at the reasons for, process of, and idealizations used in choosing a decomposition, and the broader effects of that choice.

(3) Properties may be aggregative for some decompositions and not for others, or more so for some than for others. If so, the degree of aggregativity may be used, consciously or unconsciously – but in any case quite rationally – as a criterion for choosing among decompositions: *we will tend to see aggregative or more aggregative decompositions as **natural** decompositions, and their parts as instances of **natural kinds**, because these decompositions provide simpler and less context-dependent regularities, theory, and mathematical models for the behavior they capture. These may be particularly revealing cuts on nature, but we must not make too much of this fact. We will tend to see these parts as special, and to make “nothing but” style reductionistic claims particularly for them. This is a pervasive kind of **functional localization fallacy** – a move from the claim that a decomposition is particularly powerful or revealing to the claim that the entites and forces it yields are all that matters* (Wimsatt, 1974; Bechtel and Richardson, 1992). *Such “nothing-but” claims are false or methodologically misleading if taken to suggest that one shouldn't bother to construct models or theories of the system at levels or with methods or decompositions other than that of the parts in question, or that these preferred entities are the only “real” ones* (Wimsatt, 1994).

(4) To the extent that partial aggregativity leads to greater physical or functional modularity of the parts (likely in evolving systems Wimsatt, 1981b: 141–142), it will tend to promote a consilience or robustness of parts' boundaries individuated using different properties of the system. *Greater robustness of the parts under that decomposition – a standard criterion of objecthood* (Wimsatt, 1981a) – *will also strongly contribute to the judgement that this is a decomposition into natural or real parts. Such claims are prima*

facie reasonable. (Robustness is a degree-property, so justifying these claims is not an all-or-nothing affair.) These judgements are all context-sensitive, so they still don't support "nothing-but" style claims.

(5) The four conditions all specify invariance of the system property under operations on the parts. For quantitative properties one can easily generate criteria for approximate or local aggregativity, in which variation of the system property within $\pm \varepsilon$ are tolerated for various values of ε (Wimsatt, 1986). In (1974) I used this strategy to describe different degrees of near-decomposeability or modularity in systems. Tolerances are useful theoretical tools because we use quantitative or formal qualitative frameworks as templates which nature may meet in varying degrees. For particularly adaptable frameworks which can be fitted to nature in various places in different possible ways, we may try many such mappings, looking for "best fits". (Maximizing the "coefficient of determination" or r-value of an equation in a linear regression is a simple example.) *Using "tolerances" for key qualitative concepts is a particularly useful strategy in a messy, inexact, and approximate world which has many regularities and stable patterns, but few exceptionless generalizations.*

(6) Since approximations are frequently used to produce equations of aggregative form, evaluating a property as aggregative depends upon the accuracy of the approximation under the conditions – which requires investigation. This may take two forms, with different consequences:

- (a) Emphasizing *conditions* for accuracy focuses attention upon the context-dependence of system properties, with failure of aggregativity often leading to using or studying the system under different conditions, hoping to find conditions under which its behavior is more aggregative, or other conditions where it may be qualitatively different. (Here the search simplifies conceptualization and analysis of data, and will also likely generate better experimental control – indicating conditions one should attempt to realize or avoid.) If conditions for aggregative behavior are found, we will be tempted (see (3)

above) to regard that as the **real** nature of the system, however unjustified that may ultimately be.

- (b) Emphasizing the *required* accuracy highlights our purposes and demands of our applications, adjusting demands or methods as required – using methods suited to the study of non-aggregative behavior (for greater accuracy) or weaker methods treating it as aggregative (for less). Awareness of our role in classifying the property as aggregative may make us more self-critical of our idealizations, and of possible biases in our problem-solving heuristics (Wimsatt, 1987, 1980b, 1985). Such awareness is important, but fixation on it may yield complementary errors: the view that classification of the system property is *merely* instrumental, or – even less accurately – is just conventional or socially determined. (‘Determines’ is commonly misused in such contexts – equivocating between “plays a role in determining” and “by itself, is sufficient to uniquely determine.”)

(7) One can also assess aggregativity in past and present *theories* of the relation of system and parts’ properties. In doing so, one must look not only at the final derived equations, but also at their derivation, to see if the idealizations and approximations used *assume* one or more of the conditions of aggregativity, and whether they are legitimate for the conditions at hand. One may thereby derive an aggregative model, and also come to understand its conditions of applicability. As the genetic and environmental examples treated in the longer version of this paper in my (2002) show, apparent units of aggregation can be very misleading, and the respects in which they can be treated as aggregates quite limited.

(8) One can track systematic changes in our view of the relation between system and parts’ properties, both analyzing their status in historically important disputes, and comparing their changing status in successively better theories of a phenomenon. Does this raise Hempel or Nagel’s worries (Wimsatt, 1986) – that aggregativity and emergence are really about our *knowledge* of the world, not about the world itself? But this information is *just* what a sophisticated realist needs – to see how the world changes as viewed through our theories, when our theories change. As long as you can tell one from

the other – object of study from tool of access – you’re okay, and you need to be able to vary *both* to assess and separate these effects. *This shows that in at least some cases, a standard argument for theory-relativity can be turned to the opposite conclusion – to get at the world behind the theories.* Criteria for aggregativity can be viewed either as statements about ontological relations in the world, or as tools for constructing and characterizing theories. It’s just that these are *different* uses. If one make their aims clear, there are no inescapable threats to objectivity.

(9) Earlier reductionistic theories of the behavior of a system will tend to have more simplifying assumptions, controlled variables, and assumed constancies, and predicates treated as monadic or of reduced order than later ones. With more realistic theories (with higher order relational properties and more complex interactions between parts), the degree and kinds of emergence postulated of system properties should tend to increase with increasingly detailed specification of the internal structure and environmental relations of the system in the model. *This is just the opposite of that predicted on the classical positivist model of emergence, which saw emergence disappearing with the progress of science, but appears to be just what is actually happening with recent increased interest in the study of complex systems.* Consider, for example the increased talk of holism and emergence accompanying the rising interest in non-linear dynamics (systems which violate (at least) the fourth condition for aggregativity). Though this move towards holism often goes along with a lot of loose (and sometimes misleading) anti-reductionist talk, it is clearly a confirmation of the strategy of analysis for the concept of emergence advanced here.

IV. REDUCTIONISMS AND BIASES REVISITED

I will now use this analysis to address some broader questions concerning reductionism – particularly how it is used and perceived in the context of an incomplete analysis – i.e our usual situation! The four conditions of aggregativity represent a powerful approximate and adjustable framework. In any given case one could evaluate how well each condition is met across different decompositions of

a system into parts. The better a decomposition meets these conditions, the more easily we can treat it as factoring the system into a set of modular parts having monadic, intrinsic, or context-independent properties. With particularly simple and theoretically productive decompositions, we will tend to view these parts and properties as instances of natural kinds, as robust, and to regard the system as “nothing more” than the collection of its parts. We have here turned an architectonic distinction between kinds of properties into a search heuristic for finding preferred, simple, “maximally reductionistic” decompositions of systems into parts – decompositions which lead readily to extremes of “nothing but” talk and disciplinary imperialism.

A reductive explanation of a behavior or a property of a system is one showing it to be mechanistically explicable in terms of the properties of and interactions among the parts of the system. What does this kind of explanation have to do with “nothing but” style reductionism? In principle, nothing – but in practice, they are temptingly connected – and confused. With total knowledge of a system, the two species of “reduction” are clearly distinguishable, but with increasing degrees of ignorance about a system, they come to look more and more alike! In our state of partial knowledge, it is not surprising that they should often be confused. A closer look at what we do in conditions of partial ignorance is especially important for fields and for explanatory tasks whose major questions are still “in process”, and just the kinds of judgements we should seek for limited, fallible, and error-prone scientists.

A system which is aggregative for a given decomposition is almost trivially mechanistically explicable: the parts all have the property in question, and enter into the explanation of how the system has it in the same simple way. Relationships with other parts are usually either monadic (i.e., nonexistent) or of relatively low order, and would tend to meet strong conditions of symmetry and homogeneity. The system would be relatively uninteresting: its parts would show no functional differentiation. *But none of these conditions follow from saying that the properties of a system are mechanistically explicable.* So to say that the behavior of a system is totally explicable in terms of the behavior of its parts is *not* to say that it is an aggregative function of the parts. (The inference *does*

go the other way. That might contribute to the confusion. But given the rarity of true aggregativity, even this fallacious inference doesn't explain much: there are too few occasions to invoke it.)

But now suppose that early in the investigation of a system (say, an organism) we think we know a good set of parts (say, the collection of its genes). If we don't yet know the diversity of ways in which these genes may interact with each other and with the physical conditions in the the organism (on all of the relevant size and time scales), we may tend to treat their interactions as all alike. We may do so *either* in a first-order simplified model of the system, which we simulate or analyze to explore its behavior, *or* in the "out of sight out of mind" blissful ignorance which often accompanies our view of a complex task before we have really gotten into it. In either case we are likely to overestimate how aggregative the system is, how simple it will be to understand its behavior, and to make the most simple-minded reductionistic claims about what can be learned from studying it at the lowest (or indeed *only* at the lowest!) level. (My 1979, 1980b, and 1985 provide examples of how reductionistic approaches often involve these assumptions, and how they emerge – and can cause trouble in seemingly benign applications of often effective reductionistic problem-solving heuristics.)

Thus, e.g., I think this – the aggregative character of models at early stages of a mechanistic investigation – explains (1) the characteristic oversimplifications in early claims made for the human genome project, and (2) the subsequent (necessary) broadening of the project to make it viable. (This included adding parallel comparative studies of a diversity of genomes of other species at different phylogenetic distances to determine what varies across those species groups, and to get some idea of its significance; and developmental and physiological studies at a variety of levels of organization of the expression of the genetic traits of interest). It also explains (and predicts) (3) the increasing moderation of claims for what we will learn from it. Finally, (4) if I am right, explanations which come out of it will also be far more contextual and qualified (and may involve the discovery of qualitatively new kinds of mechanisms and interactions). *This pattern – these four changes in the character of the program and the claims made for it – are not only explicable after the fact, but predictable in advance, given the normally applicable*

reductionistic research strategies and their biases (see 1980b and 1985).

This pattern is not a necessary one. There are many technologies [e.g., the computer, the automobile, or the airplane] for which early projections – indeed, projections at almost every stage – have underestimated their potential for development and their impacts on our way of life. So it is striking that we can explain this one, and how common it is.

I neglect here the obvious political purposes served by exaggerating how much could be achieved with how little. They aren't needed to explain the phenomena, which arise simply out of cognitive biases, but the phenomena are also often overdetermined, and reductionistic claims often **do** serve political ends, inside and outside of academe. (And most of the offending claimants do (or should, if they are honest) start with the disclaimer, "In principle, . . .". Yes, the road to hell *is* paved with good intentions!)¹¹ If we are to assess these claims fairly, we *must* recognize the limitations of our knowledge, the heuristic character of our tools, and the specific biases which are likely to result from their application. Such widespread knowledge earlier of how we tend to construct models and theories, and how we make claims in contexts of partial ignorance might well have produced a better project, pursued at a more reasonable pace, with more realistic ends, and at lower cost to the other fields which we will have to support and develop in any case in order to decode the texts we find in "the book of life".

But this chastening scepticism about reductionism appears to run counter to the facts. If this is true, why should reductionist methodologies have appeared to be so successful? In 1985, I noted four general properties of heuristic principles,¹² the last of which is particularly relevant here. A heuristic principle succeeds in part by transforming a problem into a different but related problem (which is easier to solve). But if it does so very effectively, there will be a strong tendency to identify the new problem as the old one – saying, "Now that we've clarified the problem so that it can be solved, . . ." or some such thing! In this way quite substantial changes in a paradigm can be hidden – particularly a cumulative string of such changes, each too small to be regarded as "fundamental". I think that this kind of *ex post facto* reification is central to the exaggeratedly high

opinion we have of reductionistic methodologies, and more generally to the largely mistaken belief that work elaborating a paradigm is merely playing out options which are already given.

But wait a minute, hasn't the reduction succeeded? Well, it has and it hasn't: its successes are often genuine, but quite misleading – it will often succeed *via* a series of subtle shifts in the question such that important aspects of the original question have not been answered. And by shifting them outside of the bounds of the new science, there will be a natural tendency to downgrade their importance. We work on problems that yield to our methods, and we all have a tendency to overestimate their power, and the importance and centrality of our own field. (It's what we know best after all!) Scientists share with others a bias which is quite general across fields and contexts.¹³ So questions that can't be addressed using our own very successful methods must be not very scientific, not very important, or both! This sounds like it's worth a good laugh, but not so fast: it may be even more dangerous than failing to solve a problem – we may now even fail to recognize that it exists!

The whole of embryology was differentiated out of the study of heredity with the rise of classical transmission genetics, between 1900 and 1926, even though developmental phenomena had figured centrally there earlier, and were regarded through much of this period as important constraints on the form of acceptable theories of heredity.¹⁴ Morgan was originally sceptical of Mendelism for that very reason – until the spectacular successes of their research program led them to simply ignore its quite paradoxical inadequacies on that score. Development again achieved the promise of a resurgence in genetics in the early 1960's with the discovery of the *lac* operon – the first account of how a gene could perform a complex and conditional control function in the expression of a trait, but a simplistic and incorrect promise it turned out to be. The genetics of elephants (or any eucaryote) was *not* like the genetics of bacteria after all – especially not the developmental genetics of metazoa. Development is now again at center stage (accelerating since the late 1970's or early 1980's) for various reasons. Important aspects of it can now be studied molecularly, and we have discovered some extremely powerful and general large-scale gene complexes (the related **HOM**, **HOX** and **DHOX** families) which

give us a handle on many more (but far from all) developmental phenomena. Is this a triumph for reductionism? In part, but it has succeeded in this by entraining and using a successively broader diversity of kinds of data and theories from other sources, and recognizing a whole new cast of causal entities. This very diversity of major players makes it much less reductionistic. The methodology (and even more, the rhetoric!) are still quite reductionistic (in the character of the heuristics used), but neither the ontology nor the epistemology are anywhere close!

NOTES

- * This paper is part of a longer one prepared originally for a festschrift for Richard Levins. The paper in that extended form will appear in my (2002). My thoughts here owe a lot to 2 writers – Ernest Nagel and Richard Levins. Some distinctions are noted by Nagel in his classic, “Wholes, Sums and Organic Unities” – though we diverge in what we make of them. Levins’ germinal essay “Complexity” (1973) and his distinction between aggregate, engineering, and evolved systems, was the immediate stimulus for this analysis and also influenced my 1974. Stuart Glennan, Peter Taylor, Bill Bechtel, Paul Humphries, and those acknowledged in my 1986 gave useful comments on prior versions.
- 1. Herbert Simon’s new essay on complexity (1996, chapter 7) adopts this view of emergence. See also Kauffman, 1992.
- 2. Wimsatt, 1976a, 1976b, 1979. For recent similar views, see Sarkar 1992, Waters 1990 and Bechtel and Richardson 1993. Some characterized by Waters (1990) as anti-reductionists are better viewed as opposing the standard philosophical analysis of reduction. (I do not cede ownership of the term to philosophers.) Most scientists see reductionism as a species of explanatory mechanistic materialism which may not even involve laws or theories, as those terms are traditionally understood by philosophers. (See the discussion of “particularistic mechanism” in Wimsatt, 1992.)
- 3. For a complementary account of the relation between causation and mechanistic explanation, see Glennan 1992.
- 4. There is a curious mis-match here: such philosophers also talk about “supervenience” – a relation between a system property and lower-level realizations of that property in which there is a many-one relationship from micro-states to that higher level property, using micro-state variables that are solely *intra*-systemic. Now called “narrow supervenience”, this is contrasted with “wide” supervenience, where the mappings depend also on the context of the system (e.g. Rosenberg (1978) on fitness). Philosophers of mind often mistakenly see “wide supervenience” as irrelevant to

their problems. Reductionistic biases (Wimsatt, 1980b) can lead us to think that narrow supervenience will do when “wide supervenience” is required. So we slip into treating fitness as a property of organisms (or even of genes), rather than as a relation between organism and environment. Moss (1992) catalogues ways in which we have attributed properties to genes that are properly characteristics of the cellular milieu or larger entities. McClamrock (1995) argues that our Cartesian heritage has misled us – seeing consciousness as an organismally or computationally “inner” property, rather than a fundamentally relational set of properties of a developed, socialized, embodied brain (Wimsatt, 1976a gives an early statement of this view).

5. As urged by Sarkar (1989, 1992) this corrects earlier analyses (1976a, 1976b), where I called an account reductionistic if there were a mechanism at any level or combination of them which explained the phenomena – even if it included mechanisms at a higher level or external to the system in question. In drawing the distinction, I would now require explicitly relativising an account to two (or more) reference levels: (1) a bottom level – the lowest level at which specific parts of a mechanism must be invoked to explain the phenomena (so an explanation of “position effect” requires characterizing an operon as a genetic control structure and how it works, but not the atomic physics accounting for the relevant electron orbitals which yield the binding which occurs.) (2) a top level – drawing the system boundaries broadly enough so that all relevant parts of the mechanisms involved are included. [Still broader things may sometimes be required, but taken for granted, as features of solar and planetary dynamics may be crucial to evolutionary processes on a variety of different time scales.] This “multi-level reductionistic analysis” picks out the appropriate levels for objects, processes, and phenomena, and explicates their relations to complete the explanatory task with no further mystery.
6. These are relevant and important criteria, but may not be completely independent, and thus fail to be separately necessary (in all combinations). They seem sufficient, but I’d welcome additional sensible requirements: each gives new handles on tough cases or special circumstances. This ‘heuristic’ attitude, first urged upon me 25 years ago by Dick Levins, makes sense – aggregativity emerges in most cases as highly conditional and qualified, with little or no foundational significance. But it may have lots of practical import: by choosing decompositions of the system to maximize their fit, these criteria can help us choose good boundaries around objects and parts of objects, acting as tools of discovery in theory *formulation and construction*. See below.
7. Some philosophers distinguish synthetic identities from realizations or instantiations. It doesn’t matter here, but see my 1994, note 30, pp. 228–229.
8. Addition, multiplication, and other operations (e.g., logical disjunction) could be appropriate in different contexts. The context of the parts properties, the system property, the question being asked, the purposes of the investiga-

tion, and the relevant applicable theories can all play important roles in such judgements.

9. This realization was brought most strikingly to my attention by Sergio Martinez in the context of a historical discussion of and argument for the robustness of primary qualities. He points out that in the 17th century something very much like aggregativity was used, together with robustness, as a criterion for natural kinds. See Martinez, 1992.
10. These approximations are endemic in the formal sciences, and in all attempts I know to build mathematical models of phenomena. Ramsey (1990) gives an analysis of the role of approximations in theory construction and justification.
11. In 1976b I discuss the purported meaning and testability of these *in principle* claims, and show that they are better seen as corollaries of identity (or localization) claims which don't make impossible demands on our computational capabilities.
12. More recently (Griesemer and Wimsatt, 1989), I have added a fifth – the explicit recognition that heuristics are purpose-relative—they are normally designed and introduced for a given purpose, which they could be expected to perform quite well. With this success, they are likely to be applied in other contexts where they do less well. Knowing the original purpose of a heuristic (or tool) may thus be very useful in determining the conditions under which (and why) it is likely to break down.
13. Abbott (2001) provides a revealing analysis of a related phenomenon. His example dealt with tendencies of the very rich and very poor to see themselves as closer to the norm than they actually were, but his mechanism seems importantly generalizable. His proposed mechanism involves tendencies to exaggerate perceived differences among positions near our own, and contracting differences among positions further away. This leads to systematic and predictable metrical distortions exaggerating the centrality/importance of one's own position. [Indeed, just such distortions characterize our visual worlds]. I urge that this is a quite general property of perspectives and of any perspectival view of the world.
14. See Allen's (1979) biography of Morgan, Mainschein's (1991) study of Wilson, Conklin, Morgan, and Harrison, Darden (1992) on early theory construction in genetics, and Wimsatt (1992) on the early history of linkage mapping.

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